How Well Do Cheap Graph Statistics Predict Random-Walk Cover Times on Small Graphs?

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Abstract

Classical upper bounds on the cover time of a simple random walk, such as $2m(n-1)$ and the cubic $\frac{4}{27}n^3$ worst case, are loose on concrete small graphs. We ask whether a handful of easily computed statistics (vertex count n, edge count m, maximum degree $d_{\max}$, diameter D) can predict the worst-start cover time of connected graphs with $n \leq 30$ more accurately than such bounds, and on which graph classes such a predictor fails. On 126 synthetic graphs from 13 families, we estimate cover times by Monte Carlo (median relative standard error 1.1%; validated against closed forms on 18 graphs) and fit log-linear predictors. A four-feature power law $\hat{C} \approx 0.56\, n^{0.83} d_{\max}^{0.70} D^{1.28}$ reaches a mean absolute log error of 0.233 [95% CI 0.195, 0.272] under *leave-one-family-out* evaluation, versus 2.19 for $2m(n-1)$ (median overestimate $8.5\times$) and 0.567 for Matthews' bound computed from exact hitting times. Diameter and maximum degree carry most of the signal; the edge count adds nothing once they are known. The predictor is reliable on paths, cycles, trees, barbells, wheels and complete bipartite graphs (within 6% on average), but fails by factors of about two on families whose behaviour the training families do not cover: lollipops and complete graphs (underestimated) and grids and dense random graphs (overestimated). It is an empirical estimate, not a bound.

1 Introduction

The cover time of a graph $G = (V, E)$ is the expected number of steps a simple random walk needs to visit every vertex. We use the worst-start version $C(G) = \max_{v \in V} \mathbb{E}_v[\tau_{\text{cov}}]$. Classical results bound $C(G) \leq 2m(n-1)$ (Aleliunas et al., 1979) and $C(G) \leq (\frac{4}{27} + o(1))n^3$ (Feige, 1995). Matthews (1988) bounds the cover time by the maximum hitting time times the harmonic number h_{n-1}; see Lovász (1993) for a survey. These bounds are tight for *some* graphs, e.g. lollipop-like graphs for the cubic bound, but on a given small graph they can be off by one to two orders of magnitude. In practice, for example when budgeting random-walk exploration or sampling on small networks, one wants a *point estimate*, not a worst-case guarantee.

Question. For connected graphs with $n \leq 30$, can a few cheap statistics give a more accurate prediction than the generic bounds, and how does the error depend on the graph type?

Contributions. (i) A reproducible Monte Carlo benchmark of 126 graphs from 13 families, with per-graph standard errors and closed-form validation. (ii) A comparison of the bounds with calibrated log-linear predictors, using both random 5-fold and *leave-one-family-out* (LOFO) cross-validation. LOFO measures transfer to an unseen graph type. (iii) A per-family error analysis and a feature ablation that show where the predictor works and where it fails. The whole study runs in about 30 s on a CPU.

2 Method

Graphs. Thirteen families, each at $n \in \{8, 12, 16, 20, 24, 30\}$. Deterministic families: path, cycle, star, complete K_n, wheel, complete bipartite $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$, 2-D grid ($2\times4, \ldots, 5\times6$), lollipop (clique $K_{n/2}$ plus a path of $n/2$ vertices), and barbell. Random families, with 3 instances per size: uniform random labelled trees; Erdős–Rényi $G(n,p)$ with $p = 2\ln n/n$ (`er-sparse`) and $p = 0.5$

Table 1: Out-of-sample accuracy over 126 graphs. $\mathrm{MAE}_{\log}$ with bootstrap 95% CI; "MC s.d." is the standard deviation of $\mathrm{MAE}_{\log}$ caused by Monte Carlo noise in the targets; "median ratio" is the median of predicted/true; the last column is the fraction of graphs predicted within $\pm 25\%$.

Predictor	$\mathrm{MAE}_{\log}$ [95% CI]	MC s.d.	median ratio	within 25%
Bound $2m(n-1)$	2.190 [1.976, 2.405]	0.0010	8.50	0%
Leading term $\frac{4}{27}n^3$	1.813 [1.649, 1.967]	0.0010	5.72	6%
Matthews $H_{\max}h_{n-1}$ (exact H)	0.567 [0.504, 0.628]	0.0010	1.75	21%
M1: $\log 2m(n-1)$ (5-fold)	0.744 [0.652, 0.842]	0.0011	1.17	25%
M1: $\log 2m(n-1)$ (LOFO)	0.815 [0.720, 0.915]	0.0012	1.22	22%
M2: $\log(mD\ln n)$ (5-fold)	0.512 [0.459, 0.567]	0.0012	1.15	22%
M2: $\log(mD\ln n)$ (LOFO)	0.567 [0.510, 0.626]	0.0013	1.14	18%
M3: $\log n, \log m$ (5-fold)	0.418 [0.348, 0.498]	0.0012	1.17	38%
M3: $\log n, \log m$ (LOFO)	0.466 [0.396, 0.547]	0.0013	1.22	26%
M4: $\log n, \log m, \log d_{\max}, \log D$ (5-fold)	0.174 [0.147, 0.203]	0.0011	0.98	76%
M4: $\log n, \log m, \log d_{\max}, \log D$ (LOFO)	0.233 [0.195, 0.272]	0.0013	0.98	64%

(`er-dense`), conditioned on connectivity; and connected random 3-regular graphs. In total there are 126 graphs, generated with NetworkX (Hagberg et al., 2008).

Monte Carlo target. For each graph, stage 1 runs $R_1 = 256$ walks from every start vertex and selects the start $v^\star$ with the largest empirical mean. Stage 2 runs $R_2 = 2000$ *fresh* walks from $v^\star$. We report their mean $\hat{C}$ and standard error $\mathrm{se} = s/\sqrt{R_2}$. Using fresh walks avoids the upward "winner's curse" bias of taking a maximum over noisy means. The only remaining bias is downward, from picking a near-worst start (quantified below).

Reference bounds. (a) $2m(n-1)$; (b) the leading term $\frac{4}{27}n^3$ of Feige's bound (for small n this is a heuristic, not a guaranteed bound); (c) Matthews' bound $H_{\max}h_{n-1}$, where the hitting times $H(u,v)$ are computed exactly by solving n linear systems. Bound (c) is not a "cheap statistic", but it is the strongest classical comparator.

Predictors. We fit ordinary least squares on $y = \log \hat{C}$ with an intercept, using four feature sets: M1: $\log 2m(n-1)$ (a calibrated version of bound (a)); M2: $\log(mD\ln n)$, motivated by resistance-based bounds; M3: $\log n, \log m$; M4: $\log n, \log m, \log d_{\max}, \log D$. The average degree $2m/n$ is a linear combination of $\log n$ and $\log m$ in log space, so M3 and M4 implicitly contain it.

Evaluation. The primary metric is the mean absolute log error $\mathrm{MAE}_{\log} = \frac{1}{N}\sum |\log \hat{C}_{\text{pred}} - \log \hat{C}|$; for example, 0.233 corresponds to a typical factor of $e^{0.233} \approx 1.26$. All fitted predictors are scored out of sample. Under 5-fold CV the folds are random; under LOFO, each family is predicted by a model fitted on the other 12 families. 95% CIs come from a nonparametric bootstrap over graphs (2000 resamples). To propagate *Monte Carlo* uncertainty, we perturb every target by $\mathcal{N}(0, (\mathrm{se}/\hat{C})^2)$, refit and re-score all models 200 times, and report the standard deviation of $\mathrm{MAE}_{\log}$ ("MC s.d."). The master seed is 2026, and per-graph RNG streams are seeded by $(2026, \text{graph id})$.

3 Experiments

Simulator validation. For paths, cycles and complete graphs, the cover time from any start is known in closed form: for the path P_n with $L = n-1$ started at vertex k it is $k(L-k)+L^2$; for the cycle it is $n(n-1)/2$; for K_n it is $(n-1)h_{n-1}$. Across these 18 graphs, the z-scores $(\hat{C} - C_{\text{exact}})/\mathrm{se}$ have mean -0.11 and maximum $|z| = 2.25$, which is consistent with $\mathcal{N}(0,1)$. Start selection is exact for cycles and complete graphs by symmetry. On paths, the selected start reaches at least 96.0% of the true worst-start cover time; this bounds the selection bias on these graphs. Over all 126 graphs, the relative standard error of $\hat{C}$ has median 1.1% and maximum 2.2%.

Main result (Table 1, Fig. 1). The generic bounds are very loose. $2m(n-1)$ overestimates by a median factor of 8.5, and the factor ranges from 1.6 (path) to 222. The cubic leading term overestimates by a median factor of 5.7; it was never below the cover time, and came closest on the n=12 lollipop (ratio 1.08). Matthews' bound is much tighter (median ratio 1.75, ratios between 0.99 and 4.0; the minimum 0.99 occurs on K_n, where the bound is exact and the shortfall is Monte

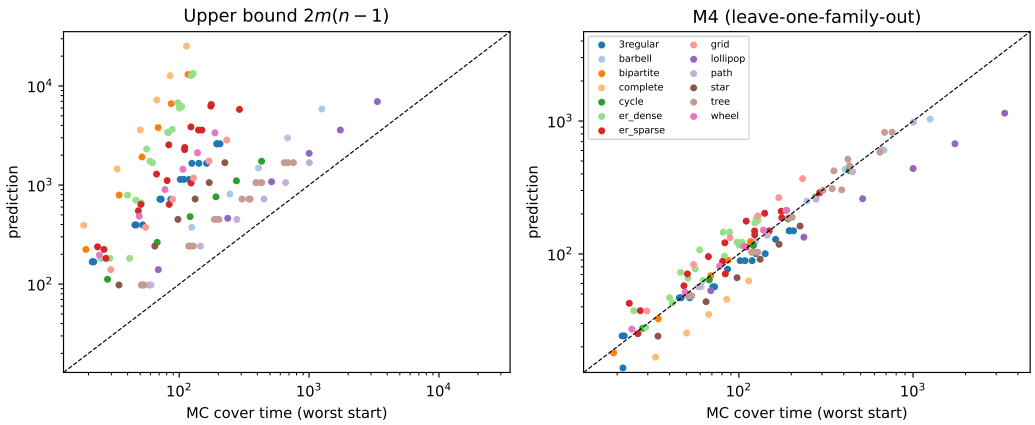


Figure 1: Predicted vs. Monte Carlo worst-start cover time (log–log). Left: bound $2m(n-1)$. Right: M4 under leave-one-family-out CV. The dashed line is $y = x$.

Table 2: Per-family geometric-mean ratio predicted/true (1.00 = unbiased). Values below 1 are underestimates.

Family	#graphs	$2m(n-1)$	Matthews	M4 5-fold	M4 LOFO
3regular	18	9.94	1.56	0.93	0.84
barbell	6	3.25	2.90	0.96	0.95
bipartite	6	40.98	1.04	0.99	0.99
complete	6	78.77	1.00	0.62	0.52
cycle	6	3.98	1.77	0.97	0.97
er-dense	18	30.41	1.68	1.20	1.32
er-sparse	18	16.68	2.02	1.17	1.21
grid	6	8.18	1.74	1.45	1.48
lollipop	6	2.05	3.29	0.64	0.48
path	6	1.64	2.73	0.96	0.94
star	6	4.82	1.03	0.83	0.69
tree	18	2.21	2.17	0.97	0.97
wheel	6	12.27	1.15	1.06	1.08

Carlo noise). Calibrating $2m(n-1)$ (M1) removes the median bias but leaves a large spread: one scale factor cannot fit both stars and paths. The four-feature model M4 has $\mathrm{MAE}_{\log}$ 0.174 [0.147, 0.203] under 5-fold CV and 0.233 [0.195, 0.272] under LOFO. Its LOFO CI does not overlap that of Matthews' bound (0.567 [0.504, 0.628]), and M4 needs no linear algebra. M4 predicts 64% of graphs within $\pm 25\%$ under LOFO; its worst-case ratios are 0.34 (underestimate) and 1.81 (overestimate). Monte Carlo noise changes $\mathrm{MAE}_{\log}$ by only about 0.001 (s.d.), two orders of magnitude below the differences between methods. Model error, not simulation error, dominates.

Fitted on all 126 graphs, M4 gives $\log \hat{C} = -0.58 + 0.83 \log n - 0.04 \log m + 0.70 \log d_{\max} + 1.28 \log D$, i.e. $\hat{C} \approx 0.56\, n^{0.83} d_{\max}^{0.70} D^{1.28}$.

Feature ablation (LOFO $\mathrm{MAE}_{\log}$, 95% CI). Full M4: 0.233 [0.195, 0.272]. Dropping $\log m$: 0.215 [0.181, 0.250]. Dropping $\log n$: 0.296 [0.253, 0.339]. Dropping $\log d_{\max}$: 0.379 [0.324, 0.439]. Dropping $\log D$: 0.521 [0.442, 0.608]. Diameter is the most important feature and maximum degree the second. The edge count is redundant given $(n, d_{\max}, D)$; dropping it slightly *improves* transfer, although the CIs overlap.

Error by graph type (Table 2, Fig. 2). Under LOFO, M4 is accurate on paths (0.94), cycles (0.97), trees (0.97), barbells (0.95), complete bipartite graphs (0.99) and wheels (1.08); the per-family mean absolute log error is ≤ 0.12 for all six. It fails in two ways. (1) *Underestimation* on lollipops (0.48), complete graphs (0.52) and stars (0.69). For complete graphs and stars, $D \leq 2$ while $d_{\max} = n - 1$, so the model sits at an extrapolated corner of feature space. For lollipops, the cover time is driven by walks that keep falling back into the clique, a mechanism that neither D nor $d_{\max}$ represents.

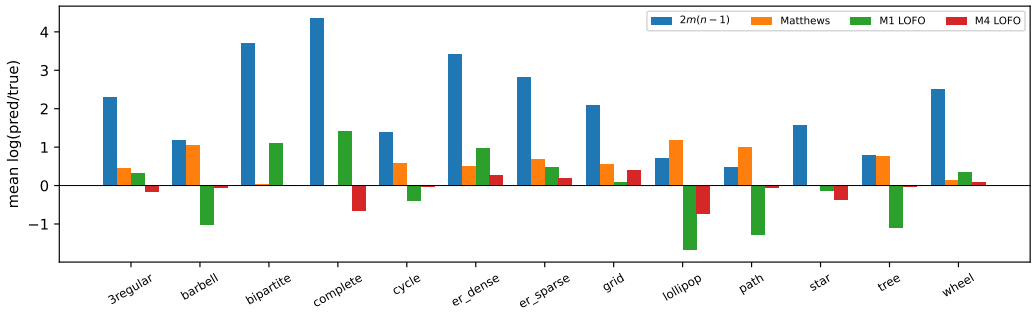


Figure 2: Mean log ratio predicted/true per family (0 = unbiased) for two bounds and two calibrated predictors (LOFO).

(2) *Overestimation* on grids (1.48), dense ER graphs (1.32) and sparse ER graphs (1.21). These are well-connected graphs whose diameter overstates how hard they are to cover. The bounds show a complementary pattern: $2m(n-1)$ is closest on paths and trees (1.6–2.2×) and worst on dense graphs (30–79×), while Matthews' bound is nearly exact on complete, star and complete bipartite graphs (ratios 1.00–1.04) and loosest on lollipops, barbells and paths (2.7–3.3×).

4 Discussion and Limitations

Answer to the question. Yes, on the graphs we tested. Four cheap statistics, effectively three since m is redundant, predict the worst-start cover time of graphs with $n \leq 30$ to within a typical factor of 1.26 on unseen families and 1.19 on unseen graphs from seen families. The bound $2m(n-1)$ is off by a typical factor of about 9, and Matthews' bound by about 1.8. The error depends strongly on the graph type: the predictor is reliable on sparse, path-like and moderately symmetric graphs, and off by about 2× on clique-plus-path structures and on the extreme dense or low-diameter corner.

Limitations. (i) *Not a bound*: M4 underestimates about half of the graphs, so it must not replace worst-case guarantees. (ii) *Distribution dependence*: the coefficients reflect our 13 families and six sizes. LOFO estimates transfer to new families, but other families (e.g. heavy-tailed or community-structured graphs) may behave differently. (iii) *Small scale*: there are 126 graphs and $n \leq 30$. We make no claim about asymptotic behaviour, and exponents fitted on a narrow range of n need not extrapolate. (iv) *Target*: we predict the worst-start cover time estimated with a two-stage start selection. On paths the selection loses at most 4%; we did not verify it exactly on non-symmetric random graphs, where the estimate is a lower bound on the true worst start up to MC noise. (v) Only linear models in log space were tried; per-family failures suggest adding a "bottleneck" feature (e.g. size of the largest clique attached to a long path), which we leave for future work. This was a deliberately small study, so no hyperparameter search or large sweep was performed.

Reproducibility. `python code/cover.py` (Python 3.11, NumPy 2.4, NetworkX 3.6; master seed 2026) regenerates all numbers, tables and figures in about 30 s on a 40-core CPU.

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